

CSE 200: Practice Online on L^AT_EX (P1)

Time: 50 Minutes

Total Marks: 50

Instructions

- Reproduce the following article using L^AT_EX exactly as it is.
- You do not need to reproduce this instruction page.
- Ensure that all text formatting, lists, nested lists, tables, equations, figures, subfigures, code listings, footnotes, hyperlinks, theorem/proposition and proof, citations, and references are implemented as presented in the article using appropriate L^AT_EX commands.
- Exact line breaks, page breaks, and float positions are not required. However, the submitted document should closely resemble the supplied article and must compile successfully.

Mark Distribution

Topic	Marks
Document Structure, Title, Authors & Affiliations	4
Lists, Nested Lists, Text Formatting & Colors	4
Fourier Series Equations & Derivations	6
Proposition & Proof (Gibbs Phenomenon)	6
Multirow and Multicolumn Table	8
Wrapped/Rotated Figure and Four Subfigures	10
Python Code Listing and Syntax Formatting	5
Cross-references	2
Footnote	1
Hyperlink	1
Citations & Bibliography	1
Overall Presentation & Successful Compilation	2
Total	50

Fourier Series and the Analysis of Periodic Signals

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1 From Signals to Series

1.1 The Fourier Series

Let $f(t)$ be a periodic function with period T . Its Fourier series representation is

$$f(t) = a_0 + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{2\pi nt}{T}\right) + b_n \sin\left(\frac{2\pi nt}{T}\right) \right], \quad (1)$$

where the coefficients are given by

$$a_0 = \frac{1}{T} \int_0^T f(t) dt, \quad a_n = \frac{2}{T} \int_0^T f(t) \cos\left(\frac{2\pi nt}{T}\right) dt, \quad b_n = \frac{2}{T} \int_0^T f(t) \sin\left(\frac{2\pi nt}{T}\right) dt. \quad (2)$$

An even function contains only cosine terms ($b_n = 0$ for all n), whereas an odd function contains only sine terms ($a_n = 0$ for all n).¹

1.2 A Simple Example: the Square Wave

For the odd square wave $f(t) = \text{sgn}(\sin(2\pi t/T))$ of amplitude 1,

$$\begin{aligned} b_n &= \frac{2}{T} \int_0^T f(t) \sin\left(\frac{2\pi nt}{T}\right) dt \\ &= \frac{4}{T} \int_0^{T/2} \sin\left(\frac{2\pi nt}{T}\right) dt \\ &= \begin{cases} \frac{4}{n\pi}, & n \text{ odd} \\ 0, & n \text{ even} \end{cases} \end{aligned} \quad (3)$$

so that the series contains only odd harmonics. A concise derivation is available from [the Square Wave entry on MathWorld](#), and the underlying convergence theory is due to Dirichlet [1].

¹A function that is neither even nor odd generally requires both families of terms.

2 Common Fourier Series Pairs

Table 1 lists several frequently used Fourier series pairs.

Family	Signal		Symmetry
	Waveform	Fundamental coefficient	
Rectangular	Square wave	$b_1 = \frac{4}{\pi}$	odd
	Pulse train (duty d)	$a_0 = d$	neither
Piecewise-linear	Triangle wave	$b_1 = \frac{8}{\pi^2}$	odd
	Sawtooth wave	$b_1 = \frac{2}{\pi}$	odd

Table 1: Selected Fourier series pairs and their leading coefficients.

3 Approximating a Square Wave from Harmonics

Figure 1 shows the ideal square wave that Equation (3) attempts to reconstruct from a finite number of harmonics. Truncating the series in Equation (1) at N terms gives the partial sum

$$s_N(t) = \sum_{\substack{n=1 \\ n \text{ odd}}}^N \frac{4}{n\pi} \sin\left(\frac{2\pi nt}{T}\right). \quad (4)$$

As N increases, $s_N(t)$ converges to $f(t)$ everywhere except at the jump discontinuities, near which an overshoot of approximately 9% persists no matter how large N becomes.

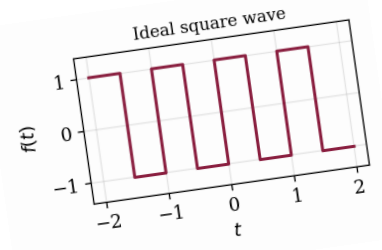


Figure 1: The ideal square

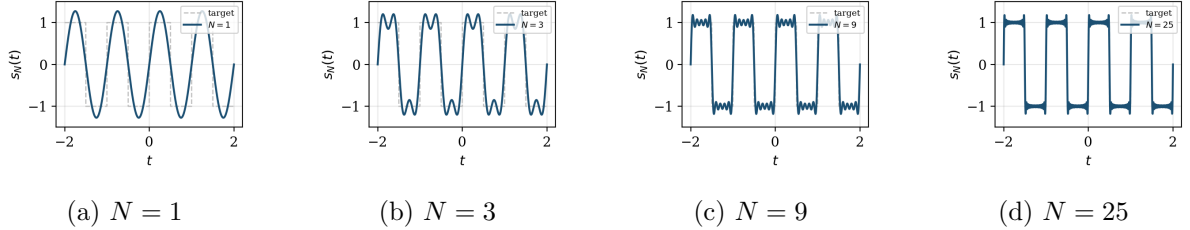


Figure 2: Partial sums $s_N(t)$ approaching the square wave as N increases. The persistent overshoot near each jump is known as the Gibbs phenomenon.

Proposition 3.1 (Gibbs Overshoot). *For the square wave partial sum $s_N(t)$ in Equation (4), the overshoot near a jump discontinuity does not vanish as $N \rightarrow \infty$; instead,*

$$\lim_{N \rightarrow \infty} s_N\left(\frac{T}{2N}\right) \approx 1.1789, \quad (5)$$

an overshoot of about 9% above the jump height of 1.

Proof. Write $s_N(t)$ as an integral of the Dirichlet kernel and evaluate at $t = T/(2N)$; the result reduces to the sine integral $\text{Si}(\pi)$, which exceeds $\pi/2$ by a fixed amount independent of N . $\square$

See Proposition 3.1 for the precise statement of this bound [2].

4 Where Fourier Analysis Is Used

Fourier analysis appears across a wide range of engineering disciplines:

- ⊕ Audio and music
 - ⊕ MP3 and AAC compression discard inaudible high-frequency content;
 - ⊕ graphic equalizers boost or cut specific frequency bands.
- ⊕ Image processing
 - ⊕ JPEG compression uses the related discrete cosine transform;
 - ⊕ edge detection filters operate on high spatial frequencies.
- ⊕ Communications
 - ⊕ modulation shifts a signal's spectrum to a carrier frequency;
 - ⊕ common multiplexing schemes include:
 - ▷ frequency-division multiplexing;
 - ▷ orthogonal frequency-division multiplexing;
 - ▷ wavelength-division multiplexing.

5 A Short Python Implementation

Listing 1 evaluates the N -term partial sum of the square wave series at a finite set of time instants using an external Python file.

Listing 1: Computing samples of a square wave partial sum.

```
import numpy as np

def square_wave_partial_sum(t, num_terms, period=1.0):
    """Return the Nth partial sum of the square wave Fourier series."""
    if num_terms <= 0:
        raise ValueError("num_terms must be positive")

    total = np.zeros_like(t)
    for k in range(1, num_terms + 1, 2):
        total += (4 / np.pi) * (1 / k) * np.sin(2 * np.pi * k * t /
            period)

    return total

t = np.linspace(-1.0, 1.0, 5)
print(square_wave_partial_sum(t, num_terms=9))
```

Equation (4) and Listing 1 describe the same computation.

References

- [1] P. G. Lejeune Dirichlet, “Sur la convergence des séries trigonométriques,” *Journal für die reine und angewandte Mathematik*, vol. 4, pp. 157–169, 1829.
- [2] A. V. Oppenheim and A. S. Willsky, *Signals and Systems*, 2nd ed. Prentice Hall, 1997.